

Power Budget: Working Document

Digital Audio Player • Rev 11

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Contents

1	Summary	2
2	Design Requirements	2
3	Method	2
4	Source Data	3
5	Conversion Efficiency	4
6	Operating Scenarios	4
6.1	S1: Standby	4
6.2	S2: Playback, display off (dominant)	5
6.3	S3: Browsing, display on	5
6.4	S4: Absolute peak (transient)	5
7	Battery Sizing	6
7.1	Selected Cell	6
7.2	Weighted Average and Derating	6
7.3	Result	6
8	Charging	7
9	Regulator and Thermal Sizing	7
10	Per-IC Current Draw	7
10.1	Load Current by IC	8
10.2	Rail Roll-Up	8
10.3	Results Against Limits	9
10.4	Nominal Current Draw (Trace Sizing)	10
11	Confidence Register	11
12	Validation Plan	11
13	Revision Log	12
14	Sources	12

1. Summary

This document converts per-block power figures into the numbers the design needs: an average draw for battery sizing, a peak draw for regulator and thermal sizing, a standby draw for shelf life, and a worst-case simultaneous draw per IC for regulator, trace and connector sizing. These are different numbers and are kept separate.

Result	Value
Playback, display off (S2)	≈ 0.94 W (254 mA at 3.7 V)
Weighted average (90% S2, 10% S3)	≈ 0.98 W (266 mA at 3.7 V)
Runtime on derated capacity	≈ 5.1 h against a 7 h target (not met)
Absolute peak (S4, Section 10)	≈ 2.51 W (837 mA at 3.0 V)
Standby (S1)	≈ 11 mW (≈ 26 days to flatten the cell)

2. Design Requirements

Requirement	Value
Battery life	≥ 7 h continuous playback on derated capacity, 90% S2 / 10% S3 usage mix
Maximum audio output	0.6 V RMS (0.85 V peak) into the Simgot EW300 (28 Ω ±15%)
Cell	Jauch LP605060JU+PCM, 1S1P, 1850 mAh (6.8 Wh)
Charge current	500 mA (0.27C)
Constraints	Low noise · Small footprint

3. Method

Summing every block's active figure produces a number no operating state reaches. Three blocks dominate that error:

- The microSD card is active roughly 2% of the time. FLAC 16/44.1 stereo is about 90 kB/s; a Class 10 card sustains 10 MB/s. Its 347 mW active figure (105 mA) contributes about 10 mW averaged.
- The OPA1622's 184 mW is full scale (0.6 V RMS) into 23.8 Ω on both channels. At IEM listening levels the output stage draws little more than quiescent. Quiescent governs battery life; peak governs thermal and regulator sizing.
- The display is off during playback by design, putting its duty cycle near 5% in the dominant use case.

Conversion losses are calculated per branch with the Section 10 method: bucks 90%, LDOs 84.6% plus ground current, LT3462A from its efficiency curve (71% at ≈ 30 mA, 67% at ≈ 100 mA). The battery is taken at 3.7 V for S1 to S3 and 3.0 V for S4.

4. Source Data

Block	Active	Passive	Rail	Basis
DAC (ES9038Q2M)	53 mW	5 mW	3.3 V	Datasheet p. 50 at DVCC = 3.3 V. 26 mW on +3V3_DAC (VCCA 2 mA, AVCC 6 mA), 26 mW on +3V3_DIG (DVCC 8 mA).
MCU (ESP32-S3)	429 mW	0.8 mW	3.3 V	Datasheet typical: 130 mA active, 240 μ A idle (ESP32-S3 Series Datasheet (Espressif) , Table 5-9)
microSD (low-speed mode)	347 mW	3 mW	3.3 V	105 mA author estimate (low-speed mode, worst case); passive typical
Oscillator (one active)	7 / 17 mW	≈ 0	3.3 V	Datasheet typical: ECS 2.2 mA / Abracon 5.2 mA; disabled part in standby
Output amp (OPA1622, 2 ch)	184 mW max	5 μ A/ch	± 3.3 V	Full scale into 23.8 Ω , both channels, plus 34.3 mW quiescent (2.6 mA $\times$ 2 per rail $\times$ 6.6 V)
I/V and summing (OPA1612, 6 ch)	143 mW	No shut-down	± 3.3 V	3.6 mA $\times$ 6 per rail $\times$ 6.6 V; off only when the analog rails are off
Display + backlight (2.4 in)	469 mW	None	3.3 V + SYS	Datasheet typical: driver 7.5 mA at 3.3 V (25 mW) + BD1604MUV 120 mA from SYS (444 mW at 3.7 V), delivering 80 mA to the LEDs
Encoder (EN12)	3.3 mW	Not given	3.3 V	Datasheet: 1 mA
Charger (BQ24250) leakage	None	60 to 80 μ W	BAT	Datasheet: 16 μ A typ, 22 μ A max, high-impedance mode

OPA1622 full-scale supply power (class AB, sine, per channel)

$$P_{\text{supply}} = \frac{2 V_s I_{\text{pk}}}{\pi} = \frac{2 \times 3.3 \text{ V} \times 35.7 \text{ mA}}{\pi} = 75 \text{ mW} \Rightarrow 2 \times 75 + 34.3 \approx 184 \text{ mW}$$

$$P_{\text{load}} = \frac{V_{\text{RMS}}^2}{R_L} = \frac{0.6^2}{23.8 \Omega} = 15 \text{ mW per channel}$$

Source: [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#) (quiescent 2.6 mA per channel, shutdown 5 μ A per channel).

Source: [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 50; [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#); [ECS-2520MVLCD Datasheet \(ECS Inc.\)](#); [ASTX-H11 Datasheet \(Abracon\)](#); [ER-TFT024IPS-3 Datasheet \(EastRising/BuyDisplay\)](#); [BD1604MUV Product Page and Datasheet \(ROHM\)](#); [EN12 Series Datasheet \(TT Electronics\)](#).

5. Conversion Efficiency

Branch	Topology	Efficiency	Note
Charger input to SYS	BQ24250, 3 MHz synchronous buck	90 to 93%	Datasheet curves at 5 V in, 500 mA
Digital 3.3 V	LTC3440 buck-boost, 600 kHz	≈ 90%	Switcher only
Analog +3.3 V	LTC3440 to 3.9 V, then LT3042	≈ 72% (S2)	90% × 84.6%, less LDO ground current
Analog -3.3 V	LTC3440 to 3.9 V, LT3462A, then LT3093	≈ 50% (S2)	Only this rail passes through the inverter. LT3462A 71% at its 31.6 mA S2 load (simulated 71.8% at 27 mA, 67.7% at 98 mA). S2 loss 96 mW: inverter 50, LT3093 27, buck 19. The LDO loss is deliberate: it buys the noise performance.
Analog combined	Both rails and +3V3_DAC	≈ 60% (S2)	394 mW from SYS delivers 236 mW
Backlight	Constant-current boost to the LED string	≈ 85%	Only when the display is on

6. Operating Scenarios

6.1 S1: Standby

Block	Draw	Basis
DAC	≤ 5 mW	Standby at DVCC = 3.3 V
MCU	0.8 mW	Idle, 240 μA
microSD	0 mW	Power-gated by load switch
Analog branch (OPA1612, OPA1622, LDOs, inverter)	≈ 0 mW	LTC3440 3.9 V shut down; the inverter follows it
Oscillators	≈ 0 mW	OE low
Display	0 mW	Off
Charger leakage	≈ 0.08 mW	16 to 22 μA from BAT in high-impedance mode
Regulator quiescent	≈ 5 mW	Estimate; sum actual I_Q for the tree
Total S1	≈ 11 mW	Sets shelf life, not runtime

At 11 mW from a 6.8 Wh cell, standby alone flattens the battery in about 26 days before cell self-discharge. This depends on power-gating the SD card and shutting the analog branch down: the OPA1612 has no shutdown pin, so its 143 mW is only removed by turning its rails off. SYSOFF is not the standby mechanism; it disconnects the battery from the system entirely and is a shipping mode.

6.2 S2: Playback, display off (dominant)

Block	Duty	Effective draw	Basis
MCU	100%	429 mW	130 mA, decode and I2S service
DAC	100%	53 mW	DVCC 8 mA + VCCA/AVCC 8 mA
microSD	2% / 98%	≈ 10 mW	$347 \times 0.02 + 3 \times 0.98$
Oscillator	100%	≈ 17 mW	Abracon (48 kHz family), 5.2 mA
OPA1622 output	100%	≈ 50 mW	34.3 mW quiescent + ≈ 16 mW typical signal
I/V and summing	100%	143 mW	Quiescent
Display	5%	≈ 23 mW	Brief wakes: $(25 + 444) \times 0.05$
Encoder	100%	3.3 mW	1 mA
Load subtotal		≈ 728 mW	
Conversion losses		≈ 210 mW	Per branch (Section 3)
Total S2		≈ 0.94 W	254 mA at 3.7 V

6.3 S3: Browsing, display on

Block	Effective draw	Basis
MCU	429 mW	Active, 130 mA
DAC + full analog chain	230 mW	DAC 53 + OPA1612 143 + OPA1622 quiescent 34; not muted during browsing
Oscillator	≈ 17 mW	5.2 mA
microSD	≈ 44 mW	12% duty during directory traversal: $347 \times 0.12 + 3 \times 0.88$
Display + backlight	469 mW	Continuous
Encoder	3.3 mW	1 mA
Conversion losses	≈ 207 mW	Per branch (Section 3)
Total S3	≈ 1.40 W	378 mA at 3.7 V

6.4 S4: Absolute peak (transient)

Every load at its maximum simultaneously; derivation in Section 10.

Block	Peak draw	Basis
MCU	429 mW	130 mA
DAC	53 mW	Active
microSD	347 mW	105 mA read burst
Oscillators	24 mW	Both counted, 7.4 mA
OPA1622 output	505 mW	34.3 mW quiescent + 71.3 mA peak per rail $\times 6.6$ V, both channels in phase
I/V and summing	143 mW	Quiescent
Display + backlight	385 mW	Driver 25 mW + LED driver 120 mA at 3.0 V
Encoder	3.3 mW	1 mA
Headroom	66 mW	5 mA per buck output, 10 mA on SYS
Conversion losses	≈ 557 mW	Per branch (Section 10)
Total S4	≈ 2.51 W	837 mA at 3.0 V

7. Battery Sizing

7.1 Selected Cell

Attribute	Value
Manufacturer / MPN	Jauch Quartz, LP605060JU+PCM+2 WIRES 70MM
DigiKey PN	1908-LP605060JU+PCM+2WIRES70MM-ND
Configuration	1S1P Li-Po, 3.7 V nominal, 3.0 to 4.2 V
Capacity	1850 mAh (6.8 Wh)
Envelope	63.0 × 51.0 × 6.6 mm (includes PCM and tab)
Termination	70 mm flying leads, 2 wires, no NTC
Lead time	28 weeks (stock available at selection)

7.2 Weighted Average and Derating

Weighted average (90% S2, 10% S3)

$$P_{\text{avg}} = 0.90 \times 938 \text{ mW} + 0.10 \times 1399 \text{ mW} = 984 \text{ mW}, \quad I_{\text{avg}} = \frac{984 \text{ mW}}{3.7 \text{ V}} = 266 \text{ mA}$$

Factor	Multiplier	Rationale
Usable depth of discharge	0.90	Cutoff above the PCM threshold so the device shuts down gracefully. The PCM trips around 2.5 V, roughly 20% below where firmware should stop, so relying on it loses power mid-write.
End-of-life capacity fade	0.85	Li-ion retains ≈ 80 to 85% of nameplate at 300 to 500 cycles. Charging at 0.27C and the switching charger's low heat both favour this figure.
Low-temperature derating	0.95	Device carried outdoors
Combined	0.73	1850 mAh × 0.73 = 1350 mAh usable

7.3 Result

Condition	Current	Runtime
Weighted average, nameplate capacity	266 mA	7.0 h
Weighted average, derated capacity (working figure)	266 mA	5.1 h
With 20% design margin, derated capacity	319 mA	4.2 h
Continuous browsing (worst realistic case)	378 mA	3.6 h

The 7 h target is not met: 5.1 h on derated capacity, the working figure (7.0 h on nameplate). The MCU (429 mW) is 59% of the S2 load and the OPA1612 (143 mW) is 20%. Reserves: MCU clock reduction to 160 MHz, or lower analog rails (the OPA1612 minimum is ±2.25 V).

8. Charging

Parameter	Value	Note
Charge current	500 mA (0.27C)	Bottom of the BQ24250's 500 mA to 2 A range; the datasheet's own design example uses this value.
Charge time	≈ 4.5 to 5 h	Including CV taper
Safety timer	9 h, set via I2C	The 6 h default leaves under an hour of margin; a required configuration step.
Charger dissipation	≈ 160 mW	At ≈ 93% efficiency
Junction rise	≈ 5 °C	θ_{JA} 32.9 °C/W (VQFN-24); regulation threshold 125 °C
Linear equivalent (rejected)	≈ 650 mW, ≈ 21 °C	Conducted into a cell 2 mm away while it nears full charge; heat plus high state of charge is the worst case for calendar aging.

- Input current is not charge current. Power is conserved through the buck: 500 mA at 5 V in yields roughly 600 mA into a 3.7 V cell at ≈ 90% efficiency.
- Concurrency is handled in hardware. PowerPath reduces charge current when system load exceeds the input limit, so the S4 peak does not collapse the rail or break termination.
- Input current limit is set by the EN1/EN2 GPIOs or over I2C. V_{IN_DPM} reduces input current to hold the supply up on a weak source or long cable.

Source: [bq2425x Product Page and Datasheet \(Texas Instruments\)](#).

9. Regulator and Thermal Sizing

These use S4 (peak), never the average. Regulator loads and margins are in Section 10.3.

Item	Result
Charger inductor	System load plus charge current plus half the ripple: 837 + 500 + 222 mA ≈ 1.56 A, against 2 A saturation. Upper bound: the 1 A input limit reduces charge current first.
OPA1622 dissipation	Worst-case class AB dissipation occurs near $0.64 \times V_{supply}$ peak output (2.1 V peak at ±3.3 V), above the 0.85 V design maximum, so the device sits on the favourable side of the curve.
SYS and BAT	Absolute maximum 5 V; everything downstream must respect this.

Load steps: SD read bursts and output-stage transients are step loads on their rails. The charger's loop changes from voltage mode in CC to current mode in CV, so SYS transient behaviour differs between charge phases.

10. Per-IC Current Draw

Two cases. Worst case (10.1 to 10.3): every load at its maximum simultaneously, battery at 3.0 V (end of discharge, highest current); used for regulator limits. Nominal (10.4): typical operation; used for trace sizing. Runtime stays on the scenario method above.

10.1 Load Current by IC

IC	Rail	Active	Idle	Basis
MCU (ESP32-S3)	+3V3_DIG	130 mA	240 μ A	Datasheet typical, Table 5-9
microSD (low-speed mode)	+3V3_DIG	105 mA	0 (gated)	Author estimate, worst case (no datasheet figure)
DAC DVCC (3.3 V logic)	+3V3_DIG	8 mA	1.2 mA	Datasheet
Display driver	+3V3_DIG	7.5 mA	Not given	Datasheet typical
Encoder (EN12)	+3V3_DIG	1 mA	Not given	Datasheet
DAC VCCA + AVCC	+3V3_DAC	2 + 6 = 8 mA	0.9 mA	Datasheet
OPA1612 (3 dual packages, 6 ch)	+3V3_ANALOG and -3V3	3.6 $\times$ 6 = 21.6 mA per rail	$\approx$ 10 μ A	Datasheet typical 3.6 mA per channel; no shutdown pin
OPA1622 quiescent (1 dual package, 2 ch)	+3V3_ANALOG and -3V3	2.6 $\times$ 2 = 5.2 mA per rail	5 μ A/ch	Datasheet typical
OPA1622 load current	+3V3_ANALOG and -3V3	71.3 mA peak per rail	0	0.6 V RMS (0.85 V peak) into 23.8 Ω , both channels in phase (LT3042 Rev 7, LT3093 Rev 10)
Oscillators (ECS + Abracon)	+3V3_ANALOG	2.2 + 5.2 = 7.4 mA	$\approx$ 0 (OE low)	Datasheet typical; both counted, disabled-part current not given
LT3042 ground current (each)	3.9 V	2.0 mA		Datasheet nominal quiescent
LT3093 ground current	-3V9	2.35 mA		LT3093 Rev 10
Backlight LED driver (BD1604MUV)	SYS	120 mA input	Not given	Datasheet typical; delivers 80 mA to the LEDs, so the LEDs are not counted again

Source: [ESP32-S3 Series Datasheet \(Espressif\)](#), Table 5-9; [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 50; [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#); [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#); [ECS-2520MVL Datasheet \(ECS Inc.\)](#); [ASTX-H11 Datasheet \(Abracon\)](#); [ER-TFT024IPS-3 Datasheet \(EastRising/BuyDisplay\)](#); [BD1604MUV Product Page and Datasheet \(ROHM\)](#); [EN12 Series Datasheet \(TT Electronics\)](#); [LT3042 Datasheet \(Analog Devices\)](#); [LT3093 Datasheet \(Analog Devices\)](#).

10.2 Rail Roll-Up

LDO input power uses 84.6% ($V_{OUT}/V_{IN} = 3.3/3.9$), so LDO input current equals output current plus ground current. Bucks use 90%. Headroom: 5 mA on each buck output, 10 mA on SYS.

+3V3_DIG (LTC3440, digital)

$$I_{OUT} = 130 + 105 + 8 + 7.5 + 1 + 5 = 256.5 \text{ mA}$$

$$I_{SYS} = \frac{3.3 \text{ V} \times 256.5 \text{ mA}}{0.90 \times 3.0 \text{ V}} = 313.5 \text{ mA}$$

+3V3_ANALOG and +3V3_DAC (LT3042 $\times$ 2)

$$I_{OUT,A} = 21.6 + 5.2 + 71.3 + 7.4 = 105.5 \text{ mA}, \quad I_{IN,A} = 105.5 + 2.0 = 107.5 \text{ mA}$$

$$I_{OUT,DAC} = 8.0 \text{ mA}, \quad I_{IN,DAC} = 8.0 + 2.0 = 10.0 \text{ mA}$$

-3V3 (LT3093) and -3V9 (LT3462A)

$$I_{OUT,LT3093} = 21.6 + 5.2 + 71.3 = 98.1 \text{ mA}, \quad I_{OUT,LT3462A} = 98.1 + 2.35 = 100.5 \text{ mA}$$

$$I_{IN,LT3462A} = \frac{3.9 \text{ V} \times 100.5 \text{ mA}}{0.67 \times 3.9 \text{ V}} = 149.9 \text{ mA}$$

Efficiency 67%: simulation gives 67.7% at 98 mA and the datasheet curve $\approx$ 69% at 100 mA (5 V in).

Source: [LT3462/LT3462A Datasheet \(Analog Devices\)](#), typical application efficiency.

3.9 V rail (LTC3440, analog)

$$I_{OUT} = 107.5 + 10.0 + 149.9 + 5 = 272.4 \text{ mA}$$

$$I_{SYS} = \frac{3.9 \text{ V} \times 272.4 \text{ mA}}{0.90 \times 3.0 \text{ V}} = 393.5 \text{ mA}$$

SYS and battery

$$I_{SYS} = 313.5 + 393.5 + 120 + 10 = 837.0 \text{ mA}, \quad P_{SYS} = 3.0 \text{ V} \times 837.0 \text{ mA} = 2.51 \text{ W}$$

10.3 Results Against Limits

Regulator / net	Worst case	Limit	Margin
LTC3440 digital output	256.5 mA	300 mA design current	1.17×
LTC3440 analog output	272.4 mA	250 mA design current	Exceeded by 9%
LT3042 (+3V3_ANALOG)	105.5 mA	125 mA programmed limit	1.18×
LT3042 (+3V3_DAC)	8.0 mA	200 mA rating	Ample
LT3093 (-3V3)	98.1 mA	125 mA programmed limit	1.27×
LT3462A switch	$\approx 0.29 \text{ A}$	300 mA min, 420 mA typ	4% under the minimum limit
SYS / BAT	837.0 mA	Cell 1850 mAh	0.45C

LT3462A switch current is the inductor average sum (149.9 + 100.5 mA) plus the 37 mA ripple from the 98 mA simulation.

10.4 Nominal Current Draw (Trace Sizing)

Typical operation with the display on: datasheet typicals, microSD at its 12% S3 duty, one oscillator, normal IEM listening (2.4 mA signal per rail), battery at 3.7 V, no headroom. Charger nets use the charging case: 500 mA charge current plus the nominal system load, charger at 90%.

IC	Rail	Nominal
MCU (ESP32-S3)	+3V3_DIG	130 mA
microSD	+3V3_DIG	13.4 mA ($105 \times 0.12 + 0.9 \times 0.88$)
DAC DVCC	+3V3_DIG	8 mA
Display driver	+3V3_DIG	7.5 mA
Encoder (EN12)	+3V3_DIG	1 mA
DAC VCCA + AVCC	+3V3_DAC	8 mA
OPA1612 (6 ch)	$\pm 3V3$	21.6 mA per rail
OPA1622 (2 ch)	$\pm 3V3$	5.2 mA quiescent + 2.4 mA signal per rail
Oscillator (one active)	+3V3_ANALOG	5.2 mA
LT3042 / LT3093 ground current	3.9 V / -3V9	2.0 mA each / 2.35 mA
Backlight LED driver (BD1604MUV)	SYS	120 mA

Net / path	Nominal	Worst case	Nominal basis
VBUS (charging)	728 mA	1.0 A	$(1424 + 1850 \text{ mW}) / 0.90 / 5 \text{ V}$
Charger SW / L → SYS (charging)	885 mA	1.56 A	385 mA system + 500 mA charge
BAT	500 mA	837 mA	Charge current (385 mA discharging)
SYS → LTC3440 digital	158.5 mA	313.5 mA	$3.3 \text{ V} \times 159.9 \text{ mA} / 0.90 / 3.7 \text{ V}$
SYS → LTC3440 analog	106.4 mA	393.5 mA	$3.9 \text{ V} \times 90.8 \text{ mA} / 0.90 / 3.7 \text{ V}$
SYS → BD1604MUV	120 mA	120 mA	Display on
+3V3_DIG	159.9 mA	256.5 mA	$130 + 13.4 + 8 + 7.5 + 1$
+3V9	90.8 mA	272.4 mA	$36.4 + 10.0 + 44.4$
+3V3_ANALOG	34.4 mA	105.5 mA	$21.6 + 5.2 + 2.4 + 5.2$
+3V3_DAC	8 mA	8 mA	VCCA + AVCC
LT3462A input (from +3V9)	44.4 mA	149.9 mA	$31.6 \text{ mA} / 0.71$
-3V9	31.6 mA	100.5 mA	$29.2 + 2.35$
-3V3	29.2 mA	98.1 mA	$21.6 + 5.2 + 2.4$
SYS total (discharging)	384.8 mA	837.0 mA	1.42 W at 3.7 V

11. Confidence Register

Item	Status	Action
Display + backlight (469 mW)	Datasheet typical	Measure the selected 2.4 in panel at operating brightness. Largest single uncertainty.
OPA1612 quiescent (3.6 mA/ch)	Datasheet typical	Confirm the maximum at ± 3.3 V; six channels at stake.
OPA1622 quiescent and shutdown	Datasheet	2.6 mA/ch typ (3.3 mA max); 5 μ A/ch shutdown.
ES9038Q2M (53 mW)	Datasheet	Confirm the power mode and oversampling setting in use.
Oscillator current maximum	Datasheet	Measure the running part.
microSD figures (105 mA)	Author estimate	Verify against an industrial card datasheet or measure with a sense resistor in the card's ground return.
SD duty cycle (2%)	Calculated	Validate by instrumenting buffer refill frequency.
Cell capacity (1850 mAh)	Datasheet	Check the discharge rate it is quoted at; the design draws $\approx 0.14C$.
Charger efficiency ($\approx 93\%$)	Datasheet curve	Verify at the actual operating point.
Conversion losses	Per branch	Bucks at an assumed 90%; LT3462A beyond its curve at peak load. Measure each branch.
Regulator quiescent (5 mW)	Estimate	Sum actual I_Q for every regulator in the tree.

12. Validation Plan

- Instrument battery current with a sense resistor in the cell return, logged over a full discharge. This supersedes every estimate here.
- Log firmware state alongside current so scenarios can be separated rather than inferred.
- Characterise the spare cell's discharge curve at operating current; the same data builds the voltage-based fuel-gauge lookup table.
- Measure a full charge cycle: current profile, time to termination, charger and cell temperature. Confirm the 9 h safety timer is never approached and thermal regulation never engages.
- Verify the TS network's actual trip temperatures; β error alone shifts the cool and warm crossings by about 3 °C.
- Measure each regulator branch on the assembled board through its test points.

13. Revision Log

Rev	Date	Change
3 to 5	Earlier	Maximum output 0.6 V RMS; OPA1622 peak 185 mW; LTC3440 at 3.9 V; negative branch changed to the LT3462A inverter; efficiency and confidence register updated.
6	2026-09-27	Reformatted. DAC 53 mW at DVCC 3.3 V; OPA1612 six channels 143 mW (arithmetic corrected); oscillator row added; OPA1622 quiescent and shutdown from the datasheet; standby corrected (analog branch shut down; OPA1612 has no shutdown); LT3462A simulated efficiency; runtime 7.8 h derated; LT3042 peak 106 mA; validation plan updated.
7	2026-10-08	Section 10 added: per-IC worst-case current draw and rail roll-up (936.7 mA, 2.81 W at 3.0 V) with results against regulator limits. ESP32-S3 datasheet linked.
8	2026-10-08	S2 to S4 recalculated from the Section 10 per-IC figures with per-branch conversion losses (loss ratios retired); source data updated; S1 MCU idle 240 μ A. S2 1.07 W, weighted average 1.12 W (302 mA), runtime 4.5 h derated: 7 h target not met. Section 9 reduced to items not covered in Section 10.
9	2026-10-08	Bases corrected: every per-IC figure except the microSD is a datasheet typical. Display, BD1604MUV and EN12 datasheets linked.
10	2026-10-08	Op-amp channel counts corrected to 6 (OPA1612) and 2 (OPA1622): 26.8 mA quiescent per rail. microSD maximum 150 $\rightarrow$ 105 mA. S2 0.94 W, weighted average 0.98 W (266 mA), runtime 5.1 h derated; S4 2.51 W (837 mA at 3.0 V); LT3462A switch $\approx$ 0.29 A.
11	2026-10-08	Section 5 split per branch for S2 (+3.3 V 72%, -3.3 V 50%, combined 60%). Section 10.4 added: nominal current per IC and per net for trace sizing (SYS 384.8 mA discharging; charger path 885 mA, VBUS 728 mA while charging).

14. Sources

1. [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#)
2. [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#)
3. [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#)
4. [ECS-2520MVLIC Datasheet \(ECS Inc.\)](#)
5. [ASTX-H11 Datasheet \(Abracon\)](#)
6. [bq2425x Product Page and Datasheet \(Texas Instruments\)](#)
7. [LTC3440 Product Page and Datasheet \(Analog Devices\)](#)
8. [LT3042 Datasheet \(Analog Devices\)](#)
9. [LT3093 Datasheet \(Analog Devices\)](#)
10. [LT3462/LT3462A Datasheet \(Analog Devices\)](#)
11. [ESP32-S3 Series Datasheet \(Espressif\)](#)
12. [ER-TFT024IPS-3 Datasheet \(EastRising/BuyDisplay\)](#)
13. [BD1604MUV Product Page and Datasheet \(ROHM\)](#)
14. [EN12 Series Datasheet \(TT Electronics\)](#)